1. Clinical Question & Core Hypothesis

Clinical Question

EMT and its downstream signaling/metabolic shifts should drive immune-evasive metastasis, yet some tumors with dense lymphoid infiltrates still fail immunotherapy. Why does immune presence not translate into therapeutic benefit in EMT-activated tumors? 2) Tier 1 paths used: the single path linking EMT → Wnt/beta-catenin → metastatic spread and the CXCR4 → GLS → cyclin D1 axis. I built on these because they form a direct mechanistic chain from EMT-associated signaling to metabolic reprogramming and cell-cycle–driven aggressiveness. 3) Tier 2 support: GLS is associated with reduced benefit from immunotherapy and is positively linked to CXCR4; cyclin D1 overexpression both characterizes aggressiveness and can be regulated by GLS, suggesting a metabolic→proliferation feedback that could suppress local immune function. 4) Why this endpoint: I chose therapy_response (immunotherapy) because the path explicitly includes reduced benefit from immunotherapy linked to GLS, making the question clinically actionable and testable using TIL imaging and gene expression. 5) Deliberate exclusions: I excluded chemotherapy agents (docetaxel/nedaplatin), drug-transporter considerations, and specific lymphoma-centric framing (MCL) to keep the hypothesis focused on the EMT→metabolism→immune-resistance mechanism in solid tumors; I limited anchor genes to GLS and CCND1 to satisfy the 2-gene cap and permit clear mechanistic testing?

Working Hypothesis

In EMT-activated solid tumors, GLS-driven glutaminolysis upregulates CCND1 expression, promoting metabolic and proliferative states that suppress tumor-infiltrating lymphocyte effector activity even when TIL density is high. Therefore, among TIL-high tumors, those with concomitantly high GLS and CCND1 expression will exhibit reduced benefit from immunotherapy compared with TIL-high tumors with low GLS and CCND1. This mechanistic axis is modulated by EMT-associated signaling and can be further influenced by secondary pathways such as CXCR4 and WNT/beta-catenin signaling, which shape GLS activity and CCND1 regulation and thereby alter immunotherapy responsiveness.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	GLS, CCND1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

GLS: GLS is associated with reduced benefit from immunotherapy and is positively linked to CXCR4; cyclin D1 overexpression both characterizes aggressiveness and can be regulated by GLS, suggesting a metabolic→proliferation feedback that could suppress local immune function.

CCND1: CCND1 to satisfy the 2-gene cap and permit clear mechanistic testing.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CXCR4	Supporting modifier from knowledge graph
Secondary	WNT/beta-catenin signaling	Supporting modifier from knowledge graph
Secondary	EMT program	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: provides a concise, testable axis for metabolic–cell cycle–driven immune resistance in EMT-activated tumors. Anchoring the hypothesis on GLS and CCND1 (two genes) with TIL density as the imaging readout enables a tractable, mechanistic investigation into why some TIL-rich tumors respond poorly to therapy.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- CXCR4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- WNT/beta-catenin signaling: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.